P.H.A.N.T.O.M.

Photonic Harassment and Neutralisation of Tactical Observation Mechanisms

Combined Technical Design Document — v3.0

Submitted by: Devang Chaturvedi — Reg. No. 23BCT0151
Submitted to: Prof. Yogesh Babu S
Programme: B.Tech Engineering Project — VIT Vellore

Version: 3.0 (incorporating LM358, ULN2003A, PCF8574)
Date: April 2026

Table of Contents

Abstract

P.H.A.N.T.O.M. (Photonic Harassment and Neutralisation of Tactical Observation Mechanisms) is a wearable, glasses-mounted system designed to autonomously detect unauthorised infrared-based imaging devices and deploy active photonic countermeasures to degrade their imaging quality. The system operates entirely on-device with no external connectivity required for its core detection-to-disruption loop. It is structured around three tightly coupled subsystems: a Sensing Layer that detects IR emissions using a dual-channel passive photodiode circuit with peak-hold pulse stretching; an Attacking Layer that fires pulsed 850 nm IR emitters and cool-white visible LEDs in a three-phase macro-cycle with pseudorandom timing; and a Power Distribution Network that isolates high-current muscle rails from the logic domain via star grounding and hierarchical decoupling. Version 3.0 integrates three new ICs over v2: the LM358 dual op-amp (unity-gain ADC buffer on all Hunter channels), the ULN2003A Darlington array (probe LED driver replacing direct GPIO drive), and the PCF8574 I2C GPIO expander (IR bank gate control, freeing ATmega328P GPIO for sensing duties). All subsystems except the Transimpedance Amplifier (TIA) front-end — which is pending faculty guidance — are implemented and bench-validated.

System Architecture Overview

P.H.A.N.T.O.M. is a wearable glasses-mounted system that autonomously detects unauthorised IR-based imaging devices and deploys active photonic countermeasures to degrade their imaging quality. The system operates entirely on-device with no external connectivity required for its core detection and response loop.

The architecture is divided into three tightly coupled layers, each with a defined role in the detection-to-disruption pipeline:

- Sensing Layer — Perceptual front-end. Detects IR emissions from cameras, LiDAR, and AF assist beams using a dual-channel passive photodiode circuit with peak-hold pulse stretching.
- Attacking Layer — Active countermeasure subsystem. Fires pulsed 850 nm IR emitters and cool-white visible LEDs in a three-phase macro-cycle with PRNG-driven timing to degrade CMOS sensor performance and disrupt autofocus and auto-exposure systems.
- Power Distribution Network — High-integrity split-rail power backbone. Isolates logic and high-current muscle rails, manages inrush, decouples noise, and governs operating modes from deep sleep through active attack.

All three layers share a common MCU: the ATmega328P. The Sensing Layer owns the ADC and detection logic. The Attacking Layer owns TIMER1/TIMER0 and the MOSFET gate drive lines. The PDN provides both stable logic supply and the high-current muscle rail for LED firing, with star grounding preventing cross-contamination between these domains.

v3.0 New ICs — Architecture Integration

IC	Layer	Role in v3.0
LM358 Dual Op-Amp	Sensing Layer	Unity-gain voltage follower buffer after each Hunter channel peak-hold capacitor. Eliminates ADC loading-induced baseline drift. Resolves the primary limitation identified in v2.
ULN2003A Darlington Array	Sensing Layer	High-current driver for the 5 × Active IR Probe LEDs. Replaces direct GPIO drive, removing 100 mA GPIO stress when all probes pulse simultaneously. Input resistors from ATmega328P GPIO; outputs drive 220 Ω + LED chains.
PCF8574 I2C GPIO Expander	Attacking Layer	8 additional quasi-bidirectional I/O pins via I2C (SDA on A4, SCL on A5). Controls IR bank gate drive lines (Bank A/B/C), freeing ATmega328P GPIO from bank-select duties and enabling cleaner pin assignment separation between sensing and attacking functions.

Effectiveness Note

> **Note:** The system is designed to degrade autofocus stability and image quality under real-world conditions. It does not claim to fully blind or permanently disable any camera system. Against PDAF-based flagship phones, three of four disruption mechanisms remain highly effective; the CDAF hunting mechanism provides partial disruption only.

System Block Diagram

Layer	Primary Hardware	MCU Interface	Function
Sensing	7× 940 nm PIN Photodiodes, BAT42, RC filters, LM358 buffers, ULN2003A probe driver	ADC0–ADC7 (8 pins)	IR detection, ratio thresholding, event confirmation
Attacking	3× OSRAM SFH4715AS, 4× 10 mm + 9× 5 mm white LEDs, IRLZ44N MOSFETs, PCF8574 expander	PD4/5/6, PB0, TIMER0/1, I2C (A4/A5)	3-phase IR saturation + visible AE disruption
Power	2S 18650, LM2596, BMS, NTC, reservoir caps	VCC, GND, star ground node	Rail isolation, inrush protection, mode management

Section 1 — Sensing Layer

1.1 Overview

The Sensing Layer is the perceptual foundation of the P.H.A.N.T.O.M. system. Its primary role is to detect the presence of active IR sources in the environment — such as camera Auto-Focus (AF) assist beams, IR illuminators, and retro-reflective glints — and distinguish them reliably from ambient solar and environmental IR noise. The layer is divided into two functional sub-systems:

- Passive IR Sensing — Photodiode-based dual-channel detection circuit (Hunter + Ambient).
- Active IR Sensing — Pulsed IR LED probes that actively illuminate the scene to trigger retro-reflective glints for detection.

1.2 Design Philosophy

1.2.1 Dual-Channel Architecture (Hunter + Ambient)

Rather than using a single photodiode and attempting to filter noise in software, the design separates detection into two independent channels:

- Hunter Channel — High-pass filtered, tuned for fast IR transients (camera pulses, glints). Six channels for spatial coverage. These are buried in 30-degree matte-black conical tubes to prevent stray light from flooding the aperture.
- Ambient Channel — Low-pass filtered, tracks the slow-moving IR baseline of the environment. It is covered in an IR-transparent frosted dome.

Detection is performed by computing the ratio between Hunter channel output and the Ambient baseline. This approach eliminates false positives caused by sunlight, halogen lighting, or other DC IR sources — the system only triggers when a fast transient exceeds the ambient level by a defined threshold.

1.2.2 Peak-Hold Pulse Stretcher

The ATmega328P ADC, even with prescaler tuning, has a minimum sample time of approximately 13 microseconds. A camera AF pulse can be as short as 5 microseconds — making direct ADC sampling of raw pulses unreliable. The solution is a passive Peak-Hold Stretcher per Hunter channel, constructed from a single BAT42 Schottky diode and a 100 nF capacitor:

- When an IR pulse hits the photodiode, photocurrent charges the 100 nF capacitor almost instantly through the low forward-voltage BAT42 diode (~0.2 V Vf).
- The diode prevents charge from leaking back, holding the peak voltage on the capacitor.
- The capacitor discharges slowly through the 10 kΩ pull-down resistor, giving a hold time of $\tau = 10\ k\Omega \times 100\ nF = 1\ ms$.
- The ADC has a full millisecond to detect the event, providing approximately 10× timing margin over the scan window.

This is an Analog Memory system. A microsecond event leaves a millisecond-scale evidence window for the MCU to read. Formally described as Passive Pulse-Width Expansion.

1.2.3 ADC Prescaler Optimisation

Instead of adding comparator ICs for faster detection, the ATmega328P ADC prescaler is tuned in firmware to increase sampling rate at the cost of minor resolution reduction:

Prescaler	ADC Clock	Sample Rate	Resolution
128 (default)	125 kHz	~9,600 sps	10-bit

Prescaler	ADC Clock	Sample Rate	Resolution
64	250 kHz	~19,200 sps	10-bit
32	500 kHz	~38,400 sps	9-bit
16 (target)	1 MHz	~76,800 sps	8-bit

At prescaler 16 with a 1 ms stretcher hold time, the system completes a full 7-channel scan cycle (~90 µs) well within the detection window. The ratio-based detection algorithm is unaffected by the reduced 8-bit resolution.

1.3 Hardware Architecture

1.3.1 Hunter Channel Signal Chain

v3.0 Update: An LM358 unity-gain op-amp buffer stage is inserted after the 100 nF hold capacitor on each Hunter channel. This eliminates ADC loading-induced baseline drift, which was identified as the primary limitation in v2. The LM358 presents near-infinite input impedance to the hold capacitor and near-zero output impedance to the ADC, ensuring stable readings across all six channels regardless of sampling order or timing.

Stage	Component	Function
Input	940 nm PIN Photodiode (Reverse Bias)	Converts IR photons to photocurrent. Reverse bias widens depletion region, reduces capacitance, improves transient response.
HPF	100 nF Ceramic (Series)	High-pass RC filter. Cutoff ~159 Hz. Blocks DC solar bias, passes fast IR transients only.
Stretcher Gate	BAT42 Schottky Diode	Low Vf (~0.2 V) allows fast charge of hold capacitor. Prevents discharge back through signal path.
Hold	100 nF Ceramic Capacitor	Peak-hold capacitor. Stores pulse voltage for ~1 ms discharge window.
Buffer (NEW v3)	LM358 Op-Amp (Unity-Gain)	Voltage follower: Vin+ = hold capacitor output, Vout = ADC input. Eliminates ADC loading. High input impedance prevents hold cap from discharging into ADC sample-and-hold circuit.
Load/Sensitivity	10 kΩ Resistor	Sets discharge time constant (τ = 1 ms). Converts photocurrent to voltage for LM358 input.
ADC Protection	1 kΩ Series Resistor	Decouples ADC sample-and-hold transients from LM358 output. Residual protection.
Output	ATmega328P ADC Pin	8-bit ADC read at prescaler 16 for maximum sample rate.

1.3.2 Ambient Channel Signal Chain

Stage	Component	Function
Input	940 nm PIN Photodiode (Reverse Bias)	Measures global environmental IR level.
LPF	10 µF Electrolytic Capacitor	Low-pass filter. Smooths all fast transients, tracks only slow baseline changes.

Stage	Component	Function
Bleeder	470 kΩ Resistor	Allows baseline to slowly adapt to changing environments. τ = 4.7 s.
ADC Protection	1 kΩ Series Resistor	Limits instantaneous current during ADC Sample-and-Hold.
Output	ATmega328P ADC Pin	Provides environmental IR reference level for ratio-based detection.

1.3.3 Dual-Path Ambient Architecture

The Ambient sensor implements a dual-path architecture allowing the MCU to distinguish between different types of environmental change. Path A (Raw) taps directly from the photodiode anode before the capacitor, responding instantaneously. Path B (Filtered), tapped after the 10 µF capacitor (τ = 4.7 s), represents the calibrated environmental baseline.

Scenario	Condition	MCU Action
Steady state	Raw ≈ Filtered	No action — system is calibrated
Walking outdoors	Raw >> Filtered	Force-charge cap via startup pulse
Shadow / indoors	Raw << Filtered	Allow 470 kΩ bleeder to drain cap slowly
Camera pulse	Raw spikes, Filtered stable	Ignore on ambient — routed to Hunter logic
Self-charged cap	Filtered high, Raw low	Suppress false trigger — MCU knows it injected the charge

1.3.4 MCU Sync / Startup Pulse Circuit

A synchronisation circuit allows the MCU to inject a known pulse into the Ambient channel at startup for self-test and calibration: MCU GPIO Pin → 220 Ω → 1N4148 Signal Diode → Ambient Capacitor. The 1N4148 acts as a check valve allowing the startup pulse to bypass the slow RC charging path without allowing the ambient capacitor to drain back into the MCU when the GPIO goes LOW.

1.3.5 Active IR Probe Sub-system

v3.0 Update: The five 940 nm IR Probe LEDs are now driven through the ULN2003A Darlington transistor array, replacing the direct GPIO drive used in v2. In v2, simultaneously pulsing all five probe LEDs at 20 mA each drew 100 mA from ATmega328P GPIO pins — within the 200 mA absolute maximum but stressful in practice. The ULN2003A accepts logic-level inputs from ATmega328P GPIO pins (each drawing < 1 mA) and drives the LED load through its open-collector Darlington outputs, which are rated to 500 mA per channel and 2.5 A total. Each probe LED retains its 220 Ω current-limiting resistor (R = (5 V − 1.2 V) / 20 mA = 190 Ω, rounded to 220 Ω standard value). Pulsed operation every 50–100 ms reduces average power consumption and extends LED lifetime.

1.3.6 TIA (Transimpedance Amplifier) Front-End — Pending

> **Status — Pending Faculty Guidance:** A Transimpedance Amplifier (TIA) front-end is planned for the Sensing Layer to provide lower-noise photocurrent-to-voltage conversion ahead of the HPF/peak-hold stage. This circuit has not yet been finalised. The author requires guidance from Prof. Yogesh Babu S to complete the TIA design (gain selection, bandwidth, op-amp selection, stability compensation). All other Sensing Layer sub-circuits are fully implemented and bench-validated. The TIA is documented here as a planned improvement; the system functions correctly without it using the existing passive signal chain.

1.4 Signal Flow Summary

Channel	Signal Path	Time Constant
Hunter (×6)	Photodiode → 100 nF HPF → BAT42 → 100 nF Hold → 10 kΩ → LM358 Buffer (NEW) → 1 kΩ → ADC	τ = 1 ms hold
Ambient (×1)	Photodiode → 10 µF LPF → 470 kΩ Bleeder → 1 kΩ → ADC	τ = 4.7 s adapt
Sync Hack	MCU GPIO → 220 Ω → 1N4148 → Ambient Cap	Startup only
IR Probes (×5)	MCU GPIO → ULN2003A Input → ULN2003A Output → 220 Ω → 940 nm IR LED (NEW)	Pulsed on demand

1.5 Software Implementation

1.5.1 Core Detection Algorithm

The core detection logic uses ratio-based thresholding rather than absolute voltage thresholds, ensuring automatic adaptation to changing lighting conditions.

- Step 1: Read all 6 Hunter channel ADC values and both Ambient ADC values (Raw + Filtered) each scan cycle.
- Step 2: Compute the Delta between Raw Ambient and Filtered Ambient to assess environmental stability.
- Step 3: If Delta is large (rapid environmental change), run the startup pulse routine and reset the confidence counter.
- Step 4: Compute the ratio between each Hunter reading and the Filtered Ambient baseline.
- Step 5: If any Hunter reading exceeds 1.5× Filtered Ambient, increment the confidence counter.
- Step 6: If confidence counter reaches 3, classify as confirmed detection and trigger the Attacking Layer via INT0 interrupt.

1.5.2 ADC Scan Cycle Timing

At prescaler 16 (target setting): single ADC conversion time ~13 µs; 7 channels per cycle (6 Hunter + 1 Filtered Ambient) = ~91 µs total scan time. Raw Ambient sampled every 3rd cycle. Stretcher hold time 1 ms provides approximately 10× margin over the scan window.

1.5.3 Confidence Counter (Temporal Confidence Integrator)

Rather than triggering on the first threshold crossing, the MCU requires three consecutive confirmations. A variable confidence is incremented each scan cycle that any Hunter reading exceeds the detection threshold; it resets to zero the moment any reading falls below threshold. At prescaler 16, three consecutive samples span approximately 40 µs. Real camera AF pulses and LiDAR returns last between 100 µs and 1 ms — long enough to build confidence. Electrical noise spikes typically last under 5 µs.

1.5.4 Dual-Threshold Hysteresis (Schmitt Trigger Behaviour)

A single trigger threshold creates chatter — rapid toggling when the signal hovers near the boundary. Two separate thresholds are used:

- Upper threshold: system activates when Hunter > 500 (8-bit ADC scale)
- Lower threshold: system deactivates only when Hunter < 400

The 100-count guard band prevents rapid toggling on the system's own reflected IR output.

1.5.5 Sensor Fusion Logic

- Spatial consistency check — At least two adjacent Hunter channels must show elevated readings for a detection event to be classified as genuine.

- Duration filter — Events resolving in under 40 µs are discarded as electrical noise. Events exceeding 5 ms are re-evaluated as sustained IR sources.
- IR probe correlation — If the Active IR Probe fired within 5 ms prior to a Hunter trigger, the system applies a reduced weight to that detection event.
- Combined decision — Only when spatial consistency, duration, confidence count, and probe correlation checks all pass does the Sensing Layer assert the detection interrupt.

1.5.6 Key Firmware Parameters

Parameter	Value	Rationale
ADC Prescaler	16	Maximises sample rate. 8-bit resolution sufficient for ratio math.
Detection Threshold	Hunter > 1.5× Filtered Ambient	Adjustable via firmware. Start conservative, tune during testing.
Confidence Requirement	3 consecutive samples	Rejects sub-40 µs noise spikes while confirming real IR pulses.
Upper Hysteresis Threshold	500 (8-bit)	Activation boundary with guard band above noise floor.
Lower Hysteresis Threshold	400 (8-bit)	Deactivation boundary — 100-count guard band below upper.
Cooldown Period	100 ms post-detection	Allows capacitors to reset and sensors to settle.
Probe LED Pulse Rate	Every 50–100 ms	Balance between detection sensitivity and power consumption.
Scan Cycle Duration	~91 µs (7 channels)	Comfortably within 1 ms stretcher window.

1.6 Bill of Materials — Sensing Layer

Qty	Component	Value / Part	Purpose
7	IR Photodiode	940 nm PIN	6× Hunter channels, 1× Ambient channel
5	IR LED	5 mm 940 nm	Active IR probes for glint/retro-reflection search
1	Op-Amp IC	LM358 Dual Op-Amp (NEW v3)	Unity-gain ADC buffer for all 6 Hunter channels
1	Darlington Driver IC	ULN2003A (NEW v3)	High-current driver for 5 IR probe LEDs, replaces direct GPIO drive
12	Ceramic Capacitor	100 nF (104)	6× HPF filter, 6× Peak-Hold stretcher
1	Electrolytic Capacitor	10 µF	Ambient LPF baseline
6	Schottky Diode	BAT42	Peak-Hold stretcher gate (one per Hunter channel)
1	Signal Diode	1N4148	Startup sync pulse routing (check valve)
6	Resistor	10 kΩ	Hunter channel discharge / sensitivity floor
7	Resistor	1 kΩ	ADC protection (6 Hunter + 1 Ambient)
6	Resistor	220 Ω	IR probe LED current limiting

Qty	Component	Value / Part	Purpose
1	Resistor	220 Ω	Startup pulse current limiter (GPIO protection)
1	Resistor	470 kΩ	Ambient channel bleeder / environment adaptation
7	Resistor	100 kΩ	Bias resistor per photodiode

1.7 Engineering Assessment — Sensing Layer

Strengths

- LM358 unity-gain buffers (v3.0 new) eliminate ADC loading — the primary limitation of v2.
- ULN2003A probe LED driver (v3.0 new) removes 100 mA GPIO stress, improving long-term reliability.
- Zero active ICs in the passive signal chain — minimal power draw and minimal failure points.
- Peak-Hold Stretcher elegantly solves the ADC speed limitation using two passive components.
- Dual-channel ratio detection is robust against environmental IR noise.
- Sensor fusion logic adds a spatial and temporal validation layer, reducing false positive rate substantially.
- Six Hunter channels provide spatial data needed for directional bank selection in the Attacking Layer.

Limitations / Pending Items

- TIA front-end not yet implemented — requires faculty guidance (Prof. Yogesh Babu S). System functions correctly without it using the existing passive signal chain.
- Pull-down resistor sensitivity is fixed in hardware — threshold tuning requires physical resistor changes or firmware compensation.
- 940 nm PIN diodes vary between manufacturers — six Hunter channels may not have perfectly matched sensitivity.
- Cannot distinguish IR source type — the system detects that IR is present, not what kind of source it is.

1.8 Academic Terminology Reference — Sensing Layer

Technique	Formal Description
Peak-Hold Stretcher	Passive Pulse-Width Expansion: A Schottky-based peak-detector array expands µs-scale IR transients into ms-scale detectable envelopes, bypassing the Nyquist limitations of the ATmega328P ADC.
LM358 Buffer (NEW v3)	Unity-Gain Voltage Follower: Op-amp configured with output shorted to inverting input, presenting near-infinite input impedance to the hold capacitor and near-zero output impedance to the ADC, eliminating sample-induced baseline drift.
ULN2003A Driver (NEW v3)	Open-Collector Darlington Array: Seven NPN Darlington pairs with integrated flyback diodes; logic-level inputs drive high-current LED loads without stressing MCU GPIO pins.
1 kΩ Series Resistor	Asymmetric Impedance Buffer: Decouples ADC sampling transients from passive filter stages, ensuring baseline stability.

Technique	Formal Description
100 nF HPF	Solar Bias Rejection: High-pass RC coupling (159 Hz cutoff) strips DC solar noise prior to the peak-detection stage.
Dual Channel Detection	Differential Ratio Thresholding: Comparing Hunter transient amplitude against the Ambient baseline enables environment-adaptive detection independent of absolute illumination level.
Confidence Counter	Temporal Confidence Integrator: Multi-sample confirmation requirement rejects sub-threshold noise spikes while validating genuine IR pulse events.
Hysteresis Band	Dual-Threshold Guard Band: Separate activation and deactivation thresholds prevent system chatter near the detection boundary, equivalent to a Schmitt Trigger.

Section 2 — Attacking Layer

2.1 Overview

The Attacking Layer is the active photonic countermeasure subsystem of the P.H.A.N.T.O.M. wearable. Its role is to degrade unauthorised imaging once the Sensing Layer has confirmed a genuine IR detection event. Activation occurs when the Sensing Layer reports a confirmed detection event (Hunter/Ambient ratio exceeding 1.5× baseline, confidence counter ≥ 3). The layer then engages a multi-modal emission sequence targeting four disruption mechanisms:

- Photodiode saturation — IR overload forces the integration capacitor beyond its reset threshold.
- Autofocus degradation — NIR pulses corrupt contrast-detection AF (CDAF) by destabilising frame-to-frame edge data. Against PDAF systems, the sub-frame jitter timing strategy degrades temporal consistency in the phase-detection pipeline.
- Auto-exposure recalibration — Visible micro-pulsed bursts collapse the sensor's dynamic range, causing convergence delays in the ISP exposure control loop.
- White balance corruption — Blue-dominant visible spectrum (5000 K cool-white) induces colour cast via Bayer filter mismatch.

Subsystem	Components	Function
Near-IR Saturation Array	3× OSRAM SFH4715AS (850 nm, 1000 mW/sr) with 20° collimating lenses	Photodiode saturation, AF disruption
Visible Spectrum Veil	4× 10 mm cool-white LEDs (5000 K) + 9× 5 mm cool-white LEDs	RGB channel overload, AE disruption via micro-pulsed flicker
PCF8574 GPIO Expander (NEW v3)	I2C expander controlling IR bank gate drive lines via single-byte write	Frees ATmega328P GPIO; single I2C write selects IR bank A/B/C gate
Passive Optical Enhancement	3M Diamond Grade DG3 retroreflective sheeting + linear polarising films (3V + 3H + 3 bare)	Signal amplification; polarised pairs exploit camera module polarisers
Power Distribution	Split-rail PDN from 2S 18650	Clean logic supply, high-current muscle rail

2.2 Design Philosophy

2.2.1 Core Strategy: Break Temporal Consistency

The fundamental goal of the Attacking Layer is not to overwhelm the camera sensor with brute force — it is to break temporal consistency faster than the ISP can stabilise. Modern PDAF systems can recover from a single saturating pulse within one frame (~33 ms at 30 fps). The system must therefore ensure disruption arrives faster than the ISP adaptation cycle, across multiple disruption domains simultaneously.

2.2.2 Three-Phase Macro-Cycle Emission Strategy

Fixed-period or uniformly random emission patterns are vulnerable to camera ISP adaptation. The v3 emission strategy uses a three-phase macro-cycle that exploits ISP adaptation lag:

- Phase 1 — Rapid Burst: 4 successive pulses at short jittered intervals (10–30 ms). This fills the ISP's temporal averaging filter with disrupted frames until it begins attempting compensation.

- Phase 2 — Deliberate Lull: A single gap of 60–75 ms (biased away from > 75 ms to prevent more than ~2 full clean frames). This allows partial ISP recalibration.
- Phase 3 — Spike: A high-intensity pulse (5 A, ≤ 150 µs) fires when the ISP has partially committed to its new estimate, causing a larger exposure error than hitting a fully-saturated sensor would. This is the false-recovery-then-hard-reset mechanism.

> **Thermal Note:** Spike cycles are non-periodic and bounded by the cumulative energy envelope rather than a fixed cycle index, preventing ISP temporal filtering from learning a periodic signature and ensuring the thermal duty limit is never exceeded.

2.2.3 Skewed PRNG Jitter Distribution

An XORSHIFT128+ pseudorandom number generator, seeded from ADC thermal noise at startup, generates emission intervals. The interval is drawn as the minimum of two uniform samples, producing a right-skewed distribution biased toward shorter intervals:

- Phase 1 (burst): intervals 10–30 ms. Sub-frame-period hits (~33 ms at 30 fps) guarantee at least one disruption per camera frame.
- Phase 2 (lull): intervals 60–75 ms. Biased away from 75–100 ms to prevent more than ~2 full clean frames.
- Phase 3 (spike): intervals 10–25 ms following the spike.
- Minimum inter-pulse recovery floor: 5 ms enforced in firmware.

2.2.4 Probabilistic Spike Injection

Spike cycles are triggered by a secondary PRNG draw with probability p ≈ 0.20, occurring during Phase 3 of the macro-cycle. This is statistically unpredictable enough that no ISP temporal filter will pattern-match it within a typical interaction window of seconds.

2.2.5 Directional Weighted Bank Selection

The six Hunter channels provide spatial information about the IR source direction. The firmware computes a weighted bank score:

- bank_A_weight = hunter[0] + hunter[1]
- bank_B_weight = hunter[2] + hunter[3]
- bank_C_weight = hunter[4] + hunter[5]

The highest-weight bank fires first in each cycle. If the second-highest bank is within 20% of the highest, both fire in rapid succession with a 1–2 ms inter-bank gap. Bank selection is locked for 2–3 pulses to prevent jittery direction switching caused by noisy Hunter readings.

2.2.6 Sequential Banked Emission and Self-Blinding Prevention

Only one SFH4715AS fires per emission event across three temple positions spaced 120° apart azimuthally. Each emitter is toed out 14–15° from the optical axis toward the adjacent bank's coverage zone, creating approximately 5° of beam overlap at coverage boundaries.

Bank	Emitter	Nominal Azimuth	Toe-out	Effective Coverage
A	SFH4715AS #1	-60° to 0°	-15°	Covers -70° to +5°
B	SFH4715AS #2	0° to +60°	+0°	Covers -5° to +65°
C	SFH4715AS #3	+60° to +120°	+15°	Covers +55° to +130°

2.2.7 Dual-Spectrum Temporal Interleaving

IR and visible emissions are interleaved rather than simultaneous:

- IR Domain (850 nm) — Targets NIR-sensitive photodiodes used in autofocus and night-vision systems. A 3 A peak current saturates the integration capacitor (Cint ≈ 1 pF) within approximately 50 µs.
- Visible Domain (5000 K cool-white) — Fires during dead-time windows between IR pulses. The blue-dominant spectrum (460 nm peak) disproportionately overloads the B channel of Bayer RGGB filters. The visible LEDs run at 1–2 kHz PWM primary flicker with occasional jumps to 4–6 kHz.

2.2.8 Polarising Film Configuration

Of the 9 emitter positions across the frame, three are fitted with vertical linear polarising film, three with horizontal linear polarising film, and three are unfiltered. Polarising films are omitted on the prototype as a deliberate tradeoff and are included as a documented upgrade path.

2.2.9 PDAF vs CDAF Effectiveness Analysis

Disruption Mechanism	CDAF Systems	PDAF Systems (v1/v2)	PDAF Systems (v3)
Photodiode Saturation (850 nm)	Highly effective	Effective	Effective + improved (20° collimation, 5 A spikes)
AF Disruption	Highly effective — frame contrast corrupted, continuous refocus loop	Partially resistant — dedicated PDAF pixels don't rely on frame contrast	Improved — sub-frame jitter degrades temporal consistency in phase-detection pipeline
AE Recalibration	Highly effective (> 500 ms convergence delay)	Effective	Effective + improved (micro-pulsed visible, macro-phase lull/spike)
White Balance Corruption	Effective	Effective	Effective

> **Honest Effectiveness Statement:** The system degrades autofocus stability and image quality under real-world conditions. It does not fully blind or permanently defeat any camera system. Against PDAF-based flagship phones, AF disruption is partial — improved by v3 timing strategy but not eliminated. AE and saturation mechanisms remain highly effective across all architectures.

2.3 Hardware Architecture

2.3.1 IR Saturation Array — Signal Chain

v3.0 Update: Bank gate select lines are now driven via PCF8574 I2C GPIO expander rather than direct ATmega328P GPIO. A single I2C byte write to the PCF8574 sets the appropriate gate drive line HIGH, triggering the IRLZ44N MOSFET for the selected bank. This frees three ATmega328P GPIO pins for other duties and cleanly separates sensing and attacking pin assignments.

Signal path: ATmega328P I2C (A4/A5) → PCF8574 → IRLZ44N Gate (Vgs = 5 V) → 2.2 Ω 5 W Cement Resistor → SFH4715AS (Vf = 1.5 V) → 20° collimating lens → anodised aluminium frame heatsink (θja = 9 K/W effective).

Parameter	Value	Notes
Wavelength	850 nm	Peak emission; NIR-sensitive photodiode target
Beam angle (with lens)	20° FWHM	Collimated from native 80°; approximately 16× irradiance gain at target
Base current	3 A	Normal pulse cycles
Spike current	5 A	Probabilistic spike cycles, ≤ 150 µs, energy-gated

Parameter	Value	Notes
Gate drive	PCF8574 I2C output → IRLZ44N (NEW v3)	IRLZ44N logic-level threshold (1–2 V Vgs(th)) ensures full saturation
Junction temperature	< 75 °C at 5% duty cycle	Maintained via frame conduction path

2.3.2 PCF8574 I2C GPIO Expander — Bank Select Interface (NEW v3)

The PCF8574 provides 8 additional quasi-bidirectional I/O pins via the ATmega328P I2C bus (SDA on A4/PC4, SCL on A5/PC5). In P.H.A.N.T.O.M., three of these outputs drive the IRLZ44N gate lines for IR Banks A, B, and C. Bank selection is performed by a single I2C write sequence:

```
// PCF8574 I2C Bank Selection — ATmega328P
// PCF8574 default I2C address: 0x20 (A0=A1=A2=GND)

#define PCF8574_ADDR  0x20
#define BANK_A_BIT    0x01   // P0
#define BANK_B_BIT    0x02   // P1
#define BANK_C_BIT    0x04   // P2

void pcf8574_write(uint8_t data) {
    Wire.beginTransmission(PCF8574_ADDR);
    Wire.write(data);
    Wire.endTransmission();
}

// Select Bank A only:
pcf8574_write(BANK_A_BIT);   // 0x01 — Bank A HIGH, B and C LOW

// Select Banks A and B simultaneously (dual-bank mode):
pcf8574_write(BANK_A_BIT | BANK_B_BIT);  // 0x03

// All banks off:
pcf8574_write(0x00);
```

2.3.3 Visible Spectrum Veil — Signal Chain

ATmega328P TIMER0 (Phase-correct PWM, 62.5 kHz base) → PB0 (4× 10 mm parallel drive + 9× 5 mm LEDs) → IRLZ44N (Ipeak = 1.4 A) → 10 Ω/1 W per LED + 47 Ω/100 nF EMI filter → Cool-White LED Array (Vf = 3.2 V each). PWM frequency is varied per dead-time window using a PRNG draw: 1–2 kHz (primary) with occasional jumps to 4–6 kHz.

2.3.4 20° Collimating Lens

Each SFH4715AS emitter is fitted with a 20° FWHM collimating lens, confirmed IR-transparent at 850 nm. The native beam angle of the SFH4715AS is 80° FWHM; collimation concentrates the same optical power into approximately 1/16th the solid angle, raising irradiance at the target by the same factor. Emitters are toed out 14–15° toward adjacent banks to create 5° beam overlap at coverage boundaries.

2.3.5 Passive Optical Integration

3M Diamond Grade DG3 Retroreflective Sheeting: ASTM D4956 Type XI, retroreflectivity RA > 550 cd/lx/m² at 0.2°/-0.2° entrance angle, full cube-corner microprism array, 0.08 mm thickness, pressure-sensitive adhesive applied to frame and temples. Note: retroreflectivity drops significantly at incidence angles exceeding 30°.

Linear Polarising Films: PVA/TAC construction, single transmission 42 ± 2% (430–780 nm), crossed extinction ≤ 0.005% (ER ≥ 8000:1). Configuration: 3 emitters with vertical film, 3 with horizontal film, 3 unfiltered.

2.3.6 Self-Blinding Prevention

Black electrical tape barriers and opaque epoxy domes physically isolate emitter zones from the Hunter photodiode apertures. The 5 ms minimum inter-pulse recovery floor in firmware prevents pulses from firing so rapidly that the Sensing Layer has insufficient time to distinguish emitted from received signals.

2.4 Software Implementation

2.4.1 Sensing Layer Trigger Interface

The Sensing Layer delivers confirmed detection events via hardware interrupt (INT0, falling edge). Conditions: hunter_max > 1.5 × ambient_filtered_mean AND ambient_raw ≈ ambient_filtered AND confidence_counter ≥ 3. Deactivation: 100 ms cooldown after last detection event.

First-pulse aggression: when detection is first confirmed, an immediate spike pulse fires before the jitter timer is drawn. This catches the camera before the ISP has stabilised its exposure estimate.

2.4.2 Emission State Machine — ISR Logic

A TIMER1 interrupt service routine running at 1 kHz implements the three-phase macro-cycle. Bank selection via PCF8574 I2C write, spike decisions, visible PWM frequency variation, and the minimum recovery floor are all PRNG-driven.

Phase	Action	Duration	Jitter Range
Phase 1 — Burst	4 successive IR pulses to facing bank (via PCF8574), visible flicker between	300 µs pulse + jitter wait	10–30 ms inter-pulse
Phase 2 — Lull	No IR emission; micro-pulsed visible fill only	Single gap	60–75 ms
Phase 3 — Spike	5 A spike (≤ 150 µs) + dual-bank if adjacent elevated + 200 ms visible burst	Spike + burst	10–25 ms post-spike
Recovery floor	Skip pulse if last pulse < 5 ms ago	Enforced each ISR tick	—
Bank lock	Lock facing bank for 2–3 pulses unless strong directional change	2–3 pulses	—

2.4.3 Directional Weighting Pseudocode

```
bank_A_weight = hunter[0] + hunter[1]
bank_B_weight = hunter[2] + hunter[3]
bank_C_weight = hunter[4] + hunter[5]
best = argmax(bank_weights)
if (bank_weights[second] > bank_weights[best] * 0.8):
    dual_bank_flag = TRUE, inter-bank gap = 1-2ms
lock bank for 2-3 pulses; re-evaluate if delta > threshold
```

2.4.4 Thermal Duty Enforcement

Firmware enforces a hard ≤ 150 µs pulse limit for spike cycles and ≤ 300 µs for base cycles, plus a 10% cumulative duty ceiling. If either limit is approached, the emission sequence pauses until the thermal interlock clears.

2.5 Emission State Machine Table

Phase	Action	Duration	Jitter Range
Phase 1 — Burst	4 successive IR pulses to facing bank (via PCF8574), visible flicker between	300 µs pulse + jitter wait	10–30 ms inter-pulse
Phase 2 — Lull	No IR emission; micro-pulsed visible fill only	Single gap	60–75 ms
Phase 3 — Spike	5 A spike (≤ 150 µs) + dual-bank if adjacent elevated + 200 ms visible burst	Spike + burst	10–25 ms post-spike
Recovery floor	Skip pulse if last pulse < 5 ms ago	Enforced each ISR tick	—
Bank lock	Lock facing bank for 2–3 pulses unless strong directional change	2–3 pulses	—

2.6 Thermal Analysis

Component	Peak Dissipation	Duty Cycle	Avg Dissipation	Margin
SFH4715AS (base, per LED)	4.5 W (3 A × 1.5 V)	5%	225 mW	Well within rated envelope
SFH4715AS (spike, per LED)	7.5 W (5 A × 1.5 V)	< 1% (≤ 150 µs)	75 mW additional	Bounded by cumulative energy envelope
2.2 Ω Cement Resistor (base)	33 W (I^2R, < 300 µs)	1%	330 mW	> 95% below trip point
2.2 Ω Cement Resistor (spike)	55 W (I^2R, ≤ 150 µs)	< 1%	82.5 mW additional	8.25 mJ per spike, within pulse rating

Maximum measured frame surface temperature during continuous 10% duty validation testing: < 55 °C.

2.7 Bill of Materials — Attacking Layer

Component	Part / Value	Supplier
IR Emitter	OSRAM SFH4715AS (850 nm)	Mouser
20° Collimating Lens	IR-transparent, confirmed 850 nm transmission	AliExpress / optical supplier
White LED (10 mm)	10 mm Ultra-Bright 5000 K	Robu.in
White LED (5 mm)	5 mm Cool-White 5000 K (9×)	Robu.in
N-Channel MOSFET	IRLZ44N	Mouser
I2C GPIO Expander (NEW v3)	PCF8574 Remote 8-Bit I/O Expander	Mouser / Amazon
Schottky Diode	1N5817	Amazon
Cement Resistor	2.2 Ω 5 W Pulse-Rated	Amazon
Retroreflective Sheeting	3M Diamond Grade DG3	Local
Polarising Film	PVA H/V, ER 10000:1	AliExpress

2.8 Engineering Assessment — Attacking Layer

Strengths

- PCF8574 I2C expander (v3.0 new) cleanly separates bank gate control from ATmega328P GPIO, freeing pins and reducing wiring complexity.
- Three-phase macro-cycle exploits ISP adaptation lag rather than relying on brute-force saturation.
- Skewed PRNG jitter with sub-frame lower bound (10 ms) guarantees at least one disruption per 30 fps camera frame cycle.
- 20° collimation provides approximately 16× irradiance gain at target.
- Directional weighted bank selection concentrates optical power toward the confirmed IR source.
- Micro-pulsed visible fill (1–6 kHz) attacks AE histogram stability.
- Thermal interlocks and cumulative energy envelope provide genuine protection against component damage.

Limitations

- PDAF autofocus is partially resistant. The v3 timing strategy improves disruption but does not eliminate PDAF's ability to recover within clean frame windows.
- The system cannot determine the direction a detected camera is pointing.
- Retroreflective sheeting effectiveness drops significantly beyond 30° incidence.
- Polarising film benefit applies primarily to the direct optical path.

2.9 Academic Terminology Reference — Attacking Layer

Technique	Formal Description
Three-Phase Macro-Cycle	Structured temporal modulation exploiting ISP adaptation lag: rapid burst phases saturate the camera pipeline's temporal filter, a deliberate lull induces partial ISP recalibration, and a subsequent high-intensity spike disrupts the sensor at its most vulnerable post-recovery state.
PCF8574 Bank Select (NEW v3)	I2C Single-Byte GPIO Write: A single Wire.write() to the PCF8574 address atomically sets the bank gate drive line, providing glitch-free bank switching without ATmega328P GPIO pin allocation.
Skewed PRNG Interval	Minimum-of-two-uniform-draws distribution: produces a right-skewed interval profile biased toward sub-frame-period disruption while retaining statistically unpredictable long gaps.
Directional Bank Prioritisation	Weighted spatial routing: Hunter channel pair sums map IR source direction to the nearest azimuthal bank, concentrating irradiance toward the confirmed source.
Micro-Pulsed Visible Fill	High-frequency luminance modulation: 1–6 kHz PWM flicker during visible emission windows induces non-stationary AE histogram instability in camera ISPs.
20° Collimation	Solid-angle concentration: collimating optics reduce beam FWHM from 80° to 20°, raising target irradiance by approximately 16× for equivalent electrical input.
10% Duty Cycle Ceiling	Thermal duty management: firmware-enforced cumulative on-time limit prevents junction temperature exceedance without requiring active cooling.

Section 3 — Power Distribution Network

3.1 Overview

The Power Distribution Network (PDN) is the backbone of the P.H.A.N.T.O.M. system. Its role is to ensure that the high-current transients generated by the Attacking Layer — up to 5 A during spike cycles — do not corrupt ADC readings in the Sensing Layer or cause MCU resets. This is achieved through strict rail separation, a hierarchical decoupling strategy, star grounding topology, and software-governed operating modes.

3.2 Split-Rail Architecture

3.2.1 Muscle Rail (7.4 V — High Power)

- Source: 2S 18650 Li-ion pack (7.4 V nominal, 8.4 V fully charged), 4000 mAh capacity
- BMS: 20 A rated — overcharge, over-discharge, short-circuit protection
- Inrush Protection: 5D-9 NTC thermistor — cold resistance ~5 Ω drops to < 0.5 Ω at operating temperature
- Reservoir: 4700 µF + 1000 µF low-ESR electrolytic capacitors (ESR < 50 mΩ)

Reservoir voltage droop — base cycle: $Q = 3\text{ A} \times 0.0003\text{ s} = 0.9\text{ mC}$; $\Delta V = 0.9\text{ mC} / 5700\ \mu\text{F} = 0.16\text{ V}$. Spike cycle: $Q = 5\text{ A} \times 0.00015\text{ s} = 0.75\text{ mC}$; $\Delta V = 0.75\text{ mC} / 5700\ \mu\text{F} = 0.13\text{ V}$. Both acceptable for the unregulated muscle rail.

3.2.2 Brain Rail (5 V — Logic and Sensing)

- Converter: LM2596 adjustable buck converter — 3 A rated, set to 5 V output. Efficiency approximately 75–80%.
- Input filter: 100 µF input capacitor at LM2596 Vin pin
- Output filter: 10 µF tantalum + 100 nF ceramic at LM2596 Vout
- MCU decoupling: 100 nF ceramic capacitor directly across ATmega328P VCC/GND pins 7 and 8
- LM358, ULN2003A, PCF8574 quiescent currents (v3.0 new): all three ICs draw from the Brain Rail. LM358 quiescent: ~1 mA typical. ULN2003A quiescent: ~0.5 mA. PCF8574 I2C idle: ~0.1 mA. Total added load: ~1.6 mA — negligible relative to the 3 A rail capacity.

3.3 Star Grounding Topology

All power return currents flow through 18 AWG heavy conductors for MOSFET and LED returns (Muscle Rail ground). All signal ground returns use 28 AWG for sensor networks, ADC reference ground, and MCU GND pin. Both converge at a single physical node: the BMS P- terminal. This prevents the $\Delta I \times Z$ noise generated by 5 A LED firing returns from flowing through the signal ground path. ADC ripple during 5 A transients is maintained below 50 mVpp, verified by oscilloscope measurement during bench validation.

> **PCB Routing Note:** In a glasses form factor with distributed PCB segments, achieving a true star ground via PCB plane pours may be impractical. Dedicated ground bus wires are required rather than plane pours, and the physical routing must be verified during assembly.

3.4 Decoupling Hierarchy

Tier	Component	Placement	Function	Frequency Range
1 — Bulk reservoir	4700 µF + 1000 µF low-ESR electrolytic	Near BMS output, on Muscle Rail	Absorbs 5 A firing transients	DC to ~1 kHz
2 — Rail filter	10 µF tantalum + 100 nF ceramic	LM2596 output, Brain Rail	Filters switching regulator ripple	1 kHz to ~500 kHz
3 — Local MCU	100 nF ceramic	Directly across ATmega328P VCC/GND pins 7 & 8	Suppresses fast switching transients at point of consumption	500 kHz to several MHz

3.5 Inrush and Protection Circuitry

3.5.1 NTC Inrush Protection

At power-on, the reservoir capacitors (5700 µF total) present a near-short-circuit to the battery. Without protection, inrush current would spike to I = V/ESR = 7.4 V / 50 mΩ = 148 A. The 5D-9 NTC thermistor limits inrush to 7.4 V / 5.05 Ω = 1.5 A during startup. After ~2 seconds, resistance drops to < 0.5 Ω.

3.5.2 BMS Protection Functions

- Overcharge protection: cuts charge above 4.2 V per cell (8.4 V total)
- Over-discharge protection: cuts discharge below 2.5 V per cell (5.0 V total)
- Short-circuit protection: < 100 µs response
- Balanced charging: ensures both cells charge to equal voltage

3.6 Operating Modes and Power Management

Mode	MCU State	Active Hardware	Avg Current	Entry / Exit Condition
1 — Deep Sleep	SLEEP_MODE_PWR_DOWN, watchdog only	Watchdog timer only	~5 mA	Entry: no detection for > 60 s. Exit: watchdog wakeup every 8 s for 91 µs scan.
2 — Active Sensing	Awake, ADC running, no emission	All 7 photodiodes, IR probes every 50–100 ms	~100 mA	Entry: watchdog wakeup or power-on. Exit: first threshold crossing.
3 — Detection Hold	Awake, ADC running, confidence tracking	Same as Active Sensing	~120 mA	Entry: confidence ≥ 1. Exit: confidence ≥ 3 → Active Attack, or drops to 0 → Active Sensing.
4 — Active Attack	Fully awake, TIMER1 ISR, 3-phase emission	IR banks, visible LED array, all sensing channels	~385 mA weighted	Entry: confirmed detection. Exit: 100 ms cooldown or thermal interlock.
5 — Thermal Cooldown	Awake, all emission forced off	Sensing channels only	~50 mA	Entry: duty cycle ceiling approached. Exit: counter resets, returns to Active Sensing.

3.7 Power Budget and Battery Life Analysis

v3.0 Update: LM358, ULN2003A, and PCF8574 quiescent currents added to Brain Rail load. Total additional draw: ~1.6 mA — negligible relative to system power budget.

Component	Peak Current	Duty Cycle	Average Current
Sensing Layer (active scan)	50 mA	100%	50 mA
IR Saturation Array (single bank, base)	3 A	5%	150 mA
IR Saturation Array (spike cycles)	5 A	< 1%	50 mA additional
Visible LED Array	1.4 A	2.5%	35 mA
MCU + Logic	50 mA	100%	50 mA
LM358 + ULN2003A + PCF8574 (NEW v3)	~1.6 mA	100%	~1.6 mA
Total (active attack mode)	—	~7.5%	~387 mA

Battery capacity: 4000 mAh (2S 18650). Continuous active attack: 4000 mAh / 387 mA ≈ 10.3 hours.

Mode	Current Draw	Estimated Time Fraction	Weighted Contribution
Deep Sleep	5 mA	50%	2.5 mA
Active Sensing	100 mA	30%	30 mA
Detection Hold	120 mA	10%	12 mA
Active Attack	387 mA	8%	30.96 mA
Thermal Cooldown	50 mA	2%	1 mA
Weighted Average	—	100%	~76.5 mA

Battery life: 4000 mAh / 76.5 mA = ~52 hours theoretical. Applying 30% derating: 52 × 0.70 ≈ 36 hours.

> **Validation Note:** The mode-time fractions above are engineering estimates based on expected deployment scenarios. Final validation data will be provided in the firmware validation report after controlled endurance testing.

3.8 Charging Architecture

Charger: Electrobot USB-C 2S Li-ion charger (CC/CV charging profile, 8.4 V maximum). Charging is performed with the BMS in-circuit. USB-C connector doubles as the programming interface for firmware uploads during development. Do not operate the system during charging.

3.9 Engineering Assessment — PDN

Strengths

- Split-rail architecture with star grounding directly addresses the primary failure mode of ADC corruption during high-current transient events.
- Hierarchical three-tier decoupling covers the full frequency range from DC bulk supply to high-frequency MCU switching transients.
- NTC inrush protection is an elegant passive solution requiring no active control circuitry.
- Five-mode software power management provides a credible, fully-published basis for the 36-hour battery life claim.

- LM358, ULN2003A, and PCF8574 (v3.0 new) add negligible quiescent current (~1.6 mA total) to the Brain Rail.

Limitations

- LM2596 efficiency is approximately 75–80%. A modern converter (TPS62xxx) would extend battery life by 5–8% and is flagged as a remaining v2/v3 improvement.
- The 36-hour figure depends on mode-time fractions requiring validation against real deployment data.
- Star ground convergence requires dedicated ground bus wires in the glasses form factor rather than PCB plane pours.

3.10 Bill of Materials — Power Distribution Network

Component	Part / Value	Supplier
2S 18650 Li-ion	4000 mAh, 2S1P	Local / Robu.in
BMS	Electronic Spices 20 A 2S	Amazon
USB-C Charger	Electrobot 2S USB-C	Amazon
NTC Thermistor	5D-9 (5 Ω cold)	Robu.in
Electrolytic Capacitor	4700 µF 16 V low-ESR	Robu.in
Electrolytic Capacitor	1000 µF 16 V low-ESR	Robu.in
Tantalum Capacitor	10 µF 16 V	Amazon
Ceramic Capacitor	100 nF 50 V	Amazon
Buck Converter	LM2596 5 V/3 A module	Amazon
LM358 Dual Op-Amp (NEW v3)	LM358N DIP-8	Mouser / Robu.in
ULN2003A Darlington Array (NEW v3)	ULN2003A DIP-16	Mouser / Amazon
PCF8574 I2C Expander (NEW v3)	PCF8574 DIP-16 or SOP-16	Mouser / Amazon
Power Wire	18 AWG silicone (Muscle Rail)	Local
Signal Wire	28 AWG (Brain Rail ground)	Local

Section 4 — Version Roadmap

4.1 v1 — Original Prototype Specification

The following represents the complete validated specification of the v1 prototype as originally submitted. All items listed here were implemented or explicitly scoped as prototype-stage constraints.

Sensing Layer — v1

- 6× Hunter photodiodes (940 nm PIN) in 30° matte-black conical tubes
- 1× Ambient photodiode with dual-path (raw + filtered) architecture
- BAT42 + 100 nF peak-hold stretcher per Hunter channel (τ = 1 ms)
- ADC prescaler 16, 8-bit resolution, ~76,800 sps
- Ratio-based detection: Hunter > 1.5× Filtered Ambient
- Confidence counter (3 consecutive samples), dual-threshold hysteresis (500/400)
- Sensor fusion: spatial consistency, duration filter, probe correlation
- 5× 940 nm IR probe LEDs pulsed every 50–100 ms
- No active op-amp buffering — accepted prototype constraint, 1 kΩ series resistors as mitigation

Attacking Layer — v1

- 3× OSRAM SFH4715AS (850 nm) with 20° collimating lenses, toed out 14–15° for 5° beam overlap
- Three-phase macro-cycle: rapid burst (Phase 1) → deliberate lull (Phase 2) → spike (Phase 3)
- Skewed PRNG jitter: min(U1, U2) distribution, 10–30 ms (Phase 1), 60–75 ms (Phase 2), 10–25 ms (Phase 3)
- Minimum 5 ms inter-pulse recovery floor
- Probabilistic spike injection: p = 0.2, 5 A, ≤ 150 µs, energy-gated
- First-pulse aggression: immediate spike on detection confirmation before jitter timer
- Weighted directional bank selection: bank_weight = hunter[n] + hunter[n+1]
- Bank lock: 2–3 pulse persistence to suppress jittery direction switching
- Direct GPIO drive of IR bank MOSFETs (replaced by PCF8574 in v3)
- 4× 10 mm + 9× 5 mm cool-white LEDs (5000 K), TIMER0 micro-pulsed at 1–6 kHz
- 3M DG3 retroreflective sheeting on frame and temples
- < 10% cumulative duty cycle, < 300 µs base pulse, ≤ 150 µs spike pulse

Power Distribution Network — v1

- 2S 18650 (4000 mAh), 20 A BMS, LM2596 5 V/3 A brain rail
- Split-rail: 7.4 V muscle + 5 V brain, star ground at BMS P-
- 4700 µF + 1000 µF reservoir, three-tier decoupling hierarchy
- 5D-9 NTC inrush protection
- Five operating modes: Deep Sleep (5 mA) → Active Sensing (100 mA) → Detection Hold (120 mA) → Active Attack (385 mA) → Thermal Cooldown (50 mA)
- 36-hour estimated battery life under mixed-use profile (50% sleep, 30% sensing, 8% active attack); requires endurance validation

4.2 v2 / v3 — Improvements and Implementation Status

The following improvements were identified in v2 planning. v3.0 marks three of them as NOW IMPLEMENTED.

Improvement	Status	Priority	Notes
Op-amp buffering on Hunter channels (LM358)	NOW IMPLEMENTED — v3.0	High	Unity-gain LM358 buffer inserted after each peak-hold capacitor. ADC loading limitation resolved.
Probe LED driver IC (ULN2003A)	NOW IMPLEMENTED — v3.0	High	ULN2003A Darlington array drives all 5 IR probe LEDs. GPIO current stress resolved.
PCF8574 I2C GPIO expander for bank select	NOW IMPLEMENTED — v3.0	High	PCF8574 controls IR bank gate drive lines via I2C, freeing ATmega328P GPIO.
TIA front-end on Hunter channels	PENDING — Faculty guidance required	High	Transimpedance amplifier front-end for lower-noise photocurrent conversion. Requires guidance from Prof. Yogesh Babu S to finalise gain, bandwidth, and stability compensation.
Replace LM2596 with TPS62xxx buck converter	Remaining v2 improvement	High	~90% efficiency vs 75–80%. Reduces heat, extends battery life by 5–8%.
Probe-assisted dynamic re-targeting	Remaining v2 improvement	Medium	Continue firing probes and reading Hunter channels during dead-time windows to update directional estimate dynamically.
State-dependent spike probability	Remaining v2 improvement	Medium	Increase spike probability when recent disruption is low, decrease when high.
Elliptical beam shaping	Remaining v2 improvement	Medium	Replace circular 20° lenses with elliptical optics for horizontal-biased camera target coverage.
Fuel gauge IC (BQ27220)	Remaining v2 improvement	Lower	Accurate battery percentage readout rather than voltage inference.
PCB layout optimisation	Remaining v2 improvement	Lower	Dedicated ground bus, separate analog/power zones, short ADC traces.
Polarising film integration	Remaining v2 improvement	Lower	Install and quantify 3V + 3H + 3 bare polarising film configuration.
NTC thermal feedback on emitters	Remaining v2 improvement	Lower	Replace theoretical duty-cycle thermal management with real NTC sensor on emitter heatsink.

Section 5 — Firmware and Validation

5.1 Overview

This section documents the hardware validation test sketches written and used during bench testing of P.H.A.N.T.O.M. subsystems. These sketches are hardware validation tools written before full firmware integration — they are not production code. Their purpose is to verify that individual hardware stages (MOSFET switching, ADC sensing, signal chain integrity) function correctly before integrating into the combined emission ISR.

5.2 Test Sketch 1 — MOSFET Gate and Zone Verification

Purpose

Verifies that the IRLZ44N MOSFETs switch correctly on each zone pin, and that the three-phase Burst → Lull → Spike pattern produces the expected waveform on each gate line. Written to test the attacking layer signal chain before integrating the full emission ISR.

Annotated Sketch

```
/*
 * P.H.A.N.T.O.M. System - Logic & MOSFET Verification
 * Testing Zones: D5, D6, D9, D10
 * Pattern: 3-Phase Attack (Burst -> Lull -> Spike)
 */

const int zones[] = {5, 6, 9, 10}; // The 4 MOSFET Gate Pins
const int numZones = 4;

void setup() {
  Serial.begin(115200);
  Serial.println("P.H.A.N.T.O.M. System Online.");
  Serial.println("Testing Zones D5, D6, D9, D10...");
  for (int i = 0; i < numZones; i++) {
    pinMode(zones[i], OUTPUT);
    digitalWrite(zones[i], LOW); // Ensure all are OFF initially
  }
}

void loop() {
  for (int z = 0; z < numZones; z++) {
    int currentPin = zones[z];
    Serial.print("Testing Zone on Pin: ");
    Serial.println(currentPin);

    // --- PHASE 1: THE BURST ---
    // Simulates filling the camera's temporal filter
    for (int i = 0; i < 4; i++) {
      digitalWrite(currentPin, HIGH);
      delay(20);
      digitalWrite(currentPin, LOW);
      delay(20);
    }

    // --- PHASE 2: THE LULL ---
    // Wait period to prevent full AGC recovery
    delay(75);

    // --- PHASE 3: THE SPIKE ---
```

```
    // The high-intensity disruption pulse
    digitalWrite(currentPin, HIGH);
    delay(15); // Visible 15ms for 5mm LED test
    digitalWrite(currentPin, LOW);

    // Short pause before moving to the next MOSFET
    delay(500);
  }
  Serial.println("All Zones Cycled. Restarting...");
  delay(1000);
}
```

Phase-by-Phase Explanation

Phase 1 — Burst (delay(20) × 4 cycles): Simulates the rapid burst phase of the full emission ISR. Four 20 ms HIGH/LOW cycles fill the ISP temporal averaging filter with disrupted frames. In the production firmware, these delays are replaced by PRNG-drawn jitter intervals of 10–30 ms.

Phase 2 — Lull (delay(75)): A fixed 75 ms pause simulates the deliberate lull phase. In this test sketch, the lull is fixed; in production firmware it is PRNG-drawn from 60–75 ms to prevent ISP pattern-matching.

Phase 3 — Spike (delay(15)): A single 15 ms HIGH pulse simulates the spike. Note: this sketch uses a 15 ms delay for safe testing with standard 5 mm LEDs. The production spike uses 5 A at ≤ 150 µs — orders of magnitude shorter and higher intensity.

Discrepancy Note — Pin Count

This sketch tests 4 zone pins (D5, D6, D9, D10). The final production specification uses 3 IR banks (D4/PD4, D5/PD5, D6/PD6 — or PCF8574 outputs in v3) plus 1 visible LED array (D8/PB0). The D9 and D10 pins in this sketch correspond to early prototype assignments that were consolidated before final pin mapping was frozen. The three-phase logic is identical to production.

Expected Serial Output and Pass Criteria

Check	Expected
Serial output	Testing Zone on Pin: 5 (then 6, 9, 10 in sequence)
Oscilloscope — Phase 1	4 × 20 ms HIGH pulses separated by 20 ms LOW gaps on the tested pin
Oscilloscope — Phase 2	75 ms LOW gap following Phase 1 burst
Oscilloscope — Phase 3	Single 15 ms HIGH pulse
Pass criterion	All four pins cycle without false triggering or cross-talk. No adjacent pin goes HIGH while the current pin is under test.

5.3 Test Sketch 2 — Sensing Layer ADC Verification

Purpose

Verifies that all Hunter photodiode channels and the Ambient channel produce correct ADC readings and respond appropriately to IR exposure. Output is formatted for the Arduino Serial Plotter to visualise all channels simultaneously.

Annotated Sketch

```
/*
 * P.H.A.N.T.O.M. Initial Test: 5 Hunters + 1 Ambient
```

```
 * Used to verify the HPF/LPF stages across all sensors.
 */

// Define Pins for the 5 Hunter Channels
const int hunterPins[] = {A0, A1, A2, A3, A4};
const int ambientPin = A5; // The reference/ambient sensor

void setup() {
  Serial.begin(115200); // High speed for real-time plotting
  for (int i = 0; i < 5; i++) {
    pinMode(hunterPins[i], INPUT);
  }
  pinMode(ambientPin, INPUT);
  Serial.println("H1,H2,H3,H4,H5,Ambient"); // Headers for the Plotter
}

void loop() {
  // Read all 5 Hunter channels
  for (int i = 0; i < 5; i++) {
    int val = analogRead(hunterPins[i]);
    Serial.print(val);
    Serial.print(",");
  }
  // Read the Ambient channel
  int ambientVal = analogRead(ambientPin);
  Serial.println(ambientVal); // Final value with println to end the frame
  delay(10); // Small delay to stabilize readings
}
```

Serial Plotter Output Explanation

When viewed in the Arduino Serial Plotter: Hunter channels (H1–H5) remain at or near the ambient noise floor under normal indoor lighting. When an IR source (camera AF assist beam, IR LED, or phone flashlight) is directed at any Hunter photodiode, that channel spikes sharply while the Ambient channel (A5/LPF) remains relatively flat due to its 4.7 s time constant. This differential behaviour confirms correct HPF/LPF filter operation.

Discrepancy Note — Channel Count

This sketch tests 5 Hunter channels (A0–A4). The final production specification uses 6 Hunter channels (A0–A5), with the 6th channel (A5) added after this sketch was written to improve spatial coverage. In production, the Ambient channel is moved to A6 (Ambient Filtered) and A7 (Ambient Raw), both of which are available as ADC-only pins on the ATmega328P that cannot be used as digital I/O — making them ideal for pure ADC inputs.

Pass Criteria

Check	Expected
Baseline reading	All channels read 20–100 (ADC counts) under ambient indoor lighting
IR exposure response	Targeted Hunter channel spikes to > 500 ADC counts when IR source is directed at it
Ambient stability	Ambient channel does not spike with fast IR pulses; changes only on sustained (> 1 s) IR exposure
Channel isolation	Non-targeted Hunter channels do not spike when one channel is illuminated

5.4 Pin Mapping Reference Table

Function	Arduino Pin	ATmega328P Port	IC Involved	TDD Reference
IR Bank A gate	D4	PD4	IRLZ44N / PCF8574 P0	Sec 2.3
IR Bank B gate	D5	PD5	IRLZ44N / PCF8574 P1	Sec 2.3
IR Bank C gate	D6	PD6	IRLZ44N / PCF8574 P2	Sec 2.3
Visible LED gate	D8	PB0	IRLZ44N	Sec 2.3
Hunter Ch 1–5	A0–A4	ADC0–ADC4	LM358 buffer output	Sec 1.3
Hunter Ch 6	A5 (production)	ADC5	LM358 buffer output	Sec 1.3
Ambient Filtered	A6	ADC6	—	Sec 1.3
Ambient Raw	A7	ADC7	—	Sec 1.3
IR Probe LEDs (×5)	D2–D7 (via ULN2003A)	Various	ULN2003A outputs	Sec 1.3
MCU Sync pulse	D3	PD3	1N4148	Sec 1.3
PCF8574 SDA	A4	PC4	PCF8574 pin 15	Sec 2.3
PCF8574 SCL	A5	PC5	PCF8574 pin 14	Sec 2.3
INT0 (detection trigger)	D2	PD2	ATmega328P hardware INT0	Sec 1.5

5.5 Validation Status Summary

Subsystem	Status	Notes
Hunter Sensing — HPF + Peak-Hold	PASS	Bench-tested with IR LED and camera AF beam
Ambient Channel — LPF + dual-path	PASS	Confirmed slow response to sustained IR; no spike on fast pulses
LM358 ADC Buffer	PASS — v3 new	Buffer inserted; ADC baseline drift eliminated vs unbuffered v2 readings
ULN2003A Probe LED Driver	PASS — v3 new	5 IR probe LEDs driven at 20 mA each; GPIO current confirmed < 1 mA per input
PCF8574 I2C Bank Select	PASS — v3 new	I2C single-byte write confirmed on oscilloscope; correct gate line goes HIGH on each bank command
MOSFET Gate Switching (3-Phase)	PASS	Verified via Test Sketch 1; all four test zones switch correctly
SFH4715AS IR Emission	PASS	IR emission confirmed with Hunter photodiode response and IR camera
Visible LED Array	PASS	1–6 kHz PWM flicker verified on oscilloscope
PDN — Star Ground + Decoupling	PASS	ADC ripple < 50 mVpp during 5 A transients (oscilloscope verified)
Battery Life (endurance)	PENDING	Mode-time fraction endurance test not yet completed

Subsystem	Status	Notes
TIA Front-End	PENDING — Faculty guidance required	TIA circuit not yet designed. Requires guidance from Prof. Yogesh Babu S on gain selection, bandwidth, and stability compensation.
Full Integrated Firmware	IN PROGRESS	ISR integration and sensor fusion logic under development

Section 6 — Emission ISR Pseudocode

6.1 Full Annotated Emission ISR Pseudocode

The following represents the complete emission state machine logic for the ATmega328P TIMER1 ISR. Pseudocode is provided for clarity; actual implementation is in C. v3.0 update: pcf8574_write() replaces direct PORT register writes for bank selection.

```
// === PHANTOM v3 Emission ISR — ATmega328P TIMER1 ===
// Runs at 1kHz. All timing in ms-equivalent tick counts.
// v3.0: Bank selection via PCF8574 I2C write

#define MIN_RECOVERY_GAP   5    // ms - minimum inter-pulse floor
#define PHASE1_PULSES      4    // rapid burst count
#define PHASE2_GAP_MIN    60    // ms
#define PHASE2_GAP_MAX    75    // ms (biased away from >75ms)
#define SPIKE_PROB        51    // out of 255 (~20%)
#define PULSE_BASE_US    300    // normal pulse width (us)
#define PULSE_SPIKE_US   150    // spike pulse width (us)

typedef enum { PHASE1_BURST, PHASE2_LULL, PHASE3_SPIKE } MacroPhase;

// PCF8574 bank selection (v3.0)
void fire_bank(uint8_t bank, uint16_t pulse_us) {
    uint8_t bank_bits[] = {0x01, 0x02, 0x04}; // Bank A/B/C
    pcf8574_write(bank_bits[bank]);
    delayMicroseconds(pulse_us);
    pcf8574_write(0x00);  // All banks off
}

// Weighted directional bank selection
uint8_t select_facing_bank(void) {
    uint16_t w[3];
    w[0] = hunter_adc[0] + hunter_adc[1]; // Bank A
    w[1] = hunter_adc[2] + hunter_adc[3]; // Bank B
    w[2] = hunter_adc[4] + hunter_adc[5]; // Bank C
    uint8_t best = argmax(w);
    uint8_t second = argmax_excluding(w, best);
    dual_bank_flag = (w[second] > (w[best] * 4) / 5);
    return best;
}

// Draw skewed jitter interval (min of two uniform draws)
uint16_t draw_interval_ms(MacroPhase phase) {
    uint8_t u1 = xorshift128_next() & 0xFF;
    uint8_t u2 = xorshift128_next() & 0xFF;
    uint8_t skewed = min(u1, u2); // biased toward low values
    if (phase == PHASE1_BURST) return 10 + (skewed * 20) / 255;
    if (phase == PHASE2_LULL)  return 60 + (skewed * 15) / 255;
    return 10 + (skewed * 15) / 255; // PHASE3_SPIKE
}

ISR(TIMER1_COMPA_vect) {
    ticks_since_last_pulse++;
    if (ticks_until_next > 0) {
        ticks_until_next--;
        run_visible_pwm_flicker(); // micro-pulsed during wait
        return;
    }
    if (ticks_since_last_pulse < MIN_RECOVERY_GAP) {
        ticks_until_next = 1; return; // enforce recovery floor
```

```
    }
    if (cumulative_duty_exceeded()) {
        enter_thermal_cooldown(); return;
    }
    uint8_t facing = select_facing_bank();
    switch (current_phase) {
    case PHASE1_BURST:
        fire_bank(facing, PULSE_BASE_US);  // PCF8574 write (v3)
        if (dual_bank_flag) {
            delay_ms(1); // 1-2ms inter-bank gap
            fire_bank(adjacent_bank, PULSE_BASE_US);
        }
        ticks_since_last_pulse = 0;
        phase1_count++;
        if (phase1_count >= PHASE1_PULSES) {
            phase1_count = 0;
            current_phase = PHASE2_LULL;
        }
        ticks_until_next = draw_interval_ms(PHASE1_BURST);
        break;
    case PHASE2_LULL:
        // No IR emission - visible fill runs in wait handler
        current_phase = PHASE3_SPIKE;
        ticks_until_next = draw_interval_ms(PHASE2_LULL);
        break;
    case PHASE3_SPIKE:
        fire_bank(facing, PULSE_SPIKE_US); // 5A spike via PCF8574
        fire_visible_burst(200);            // 200ms high-intensity burst
        ticks_since_last_pulse = 0;
        current_phase = PHASE1_BURST;
        ticks_until_next = draw_interval_ms(PHASE3_SPIKE);
        break;
    }
}

void run_visible_pwm_flicker(void) {
    uint8_t r = xorshift128_next() & 0x0F;
    OCR0A = (r < 12) ? 124 : 31; // 1-2kHz or 4-6kHz
}
```

6.2 Key Firmware Parameters — Attacking Layer

Parameter	Value	Rationale
Phase 1 burst count	4 pulses	Fills ISP temporal filter before lull
Phase 1 interval	10–30 ms (skewed)	Sub-frame lower bound guarantees ≥ 1 hit per 33 ms camera frame
Phase 2 lull	60–75 ms	Allows partial ISP recalibration without granting full clean-frame recovery
Phase 3 interval	10–25 ms post-spike	Quick follow-up while sensor is in post-spike state
Min recovery floor	5 ms	Avoids wasted pulses on fully-saturated integration capacitors
Spike probability	p = 0.20	Statistically unpredictable within typical interaction window

Parameter	Value	Rationale
Spike pulse width	≤ 150 µs	Short enough to stay within thermal budget even at 5 A
Base pulse width	≤ 300 µs	Sufficient for integration capacitor overflow at 3 A
Bank lock duration	2–3 pulses	Suppresses jittery direction switching from noisy Hunter readings
Dual-bank threshold	Second-best within 20% of best	Fires both banks when camera is between coverage zones
Bank select method	PCF8574 I2C single-byte write (v3)	Replaces direct GPIO PORT write; cleaner separation of pin domains
Visible PWM primary	1–2 kHz	Multiple transitions within 10–30 ms AE integration window
Visible PWM secondary	4–6 kHz (occasional)	Adds unpredictability to the visible disruption signature
PRNG seed	ADC thermal noise at startup	Non-deterministic seed; PRNG state stored in EEPROM for replay validation

Section 7 — References

[1] OSRAM Opto Semiconductors. SFH4715AS High Power Infrared Emitter (850 nm) Datasheet. OSRAM GmbH, 2021.

[2] Infineon Technologies. IRLZ44N Logic Level N-Channel Power MOSFET Datasheet. Infineon Technologies AG, 2004.

[3] Microchip Technology. ATmega328P 8-bit AVR Microcontroller Datasheet. Microchip Technology Inc., 2015.

[4] Texas Instruments. LM2596 SIMPLE SWITCHER Power Converter 150 kHz 3A Step-Down Voltage Regulator Datasheet. Texas Instruments Inc., 2020.

[5] Texas Instruments. LM358 Dual Operational Amplifier Datasheet. Texas Instruments Inc., 2023.

[6] Texas Instruments. ULN2003A High-Voltage, High-Current Darlington Transistor Arrays Datasheet. Texas Instruments Inc., 2015.

[7] Texas Instruments. PCF8574 Remote 8-Bit I/O Expander for I2C Bus Datasheet. Texas Instruments Inc., 2015.

[8] STMicroelectronics. BAT42 Small Signal Schottky Diode Datasheet. STMicroelectronics, 2016.

[9] Hamamatsu Photonics. Si PIN Photodiode S6468 Series (940 nm) Datasheet. Hamamatsu Photonics K.K., 2020.

[10] 3M Traffic Safety Systems Division. Diamond Grade DG3 Reflective Sheeting Series — Product Specification (ASTM D4956 Type XI). 3M Company, 2018.

[11] Vigna, S. (2017). Further scramblings of Marsaglia's xorshift generators. Journal of Computational and Applied Mathematics, 315, 175–181.

[12] Rukhin, A., Soto, J., Nechvatal, J., et al. A Statistical Test Suite for Random and Pseudorandom Number Generators for Cryptographic Applications. NIST Special Publication 800-22 Rev. 1a. National Institute of Standards and Technology, 2010.

[13] Ott, H. W. (2009). Electromagnetic Compatibility Engineering. John Wiley & Sons. (Chapters 4–5: Star grounding, mixed-signal PCB layout.)

[14] Fossum, E. R. and Hondongwa, D. B. (2014). A Review of the Pinned Photodiode for CCD and CMOS Image Sensors. IEEE Journal of the Electron Devices Society, 2(3), 33–43. (Bayer filter, CMOS photodiode integration, ISP pipeline.)

[15] El-Anwar, M. and Kamel, T. S. (2014). A Comparative Study of Contrast Detection Autofocus versus Phase Detection Autofocus in Digital Cameras. IEEE International Conference on Computer Engineering and Systems (ICCES). (CDAF vs PDAF autofocus architecture.)

Appendix A — Consolidated Bill of Materials

All three layers merged into a single procurement table.

Layer	Component	Qty	Purpose	Supplier
SENSING	IR Photodiode 940 nm PIN	7	Hunter channels (6) + Ambient (1)	Robu.in / AliExpress
SENSING	5 mm IR LED 940 nm	5	Active IR probe LEDs	Robu.in
SENSING	LM358 Dual Op-Amp (NEW v3)	1	Unity-gain ADC buffer for Hunter channels	Mouser / Robu.in
SENSING	ULN2003A Darlington Array (NEW v3)	1	Probe LED high-current driver	Mouser / Amazon
SENSING	BAT42 Schottky Diode	6	Peak-hold stretcher gate	Mouser
SENSING	1N4148 Signal Diode	1	Startup sync pulse check valve	Robu.in
SENSING	100 nF Ceramic Capacitor	12	HPF (6) + peak-hold (6)	Amazon
SENSING	10 µF Electrolytic Capacitor	1	Ambient LPF	Robu.in
SENSING	10 kΩ Resistor	6	Hunter discharge / sensitivity	Amazon
SENSING	1 kΩ Resistor	7	ADC protection	Amazon
SENSING	470 kΩ Resistor	1	Ambient bleeder	Amazon
SENSING	220 Ω Resistor	7	Probe LED current limit (5) + startup (1) + misc (1)	Amazon
SENSING	100 kΩ Resistor	7	Photodiode bias	Amazon
ATTACKING	OSRAM SFH4715AS (850 nm)	3	IR saturation emitters	Mouser
ATTACKING	20° Collimating Lens	3	IR beam collimation	AliExpress
ATTACKING	10 mm Cool-White LED 5000 K	4	Visible spectrum veil	Robu.in
ATTACKING	5 mm Cool-White LED 5000 K	9	Visible spectrum veil	Robu.in
ATTACKING	IRLZ44N N-Channel MOSFET	4	Gate drive for IR banks + visible array	Mouser
ATTACKING	PCF8574 I2C GPIO Expander (NEW v3)	1	IR bank gate select control	Mouser / Amazon
ATTACKING	2.2 Ω 5 W Cement Resistor	3	IR bank current limit	Amazon
ATTACKING	3M DG3 Retroreflective Sheeting	1 sheet	Passive optical enhancement	Local
ATTACKING	Linear Polarising Film (H/V pair)	2 sets	Optional polarisation path (v2 upgrade)	AliExpress
PDN	2S 18650 Li-ion 4000 mAh	1 pack	Main battery	Robu.in / Local
PDN	20 A 2S BMS	1	Battery protection and balancing	Amazon
PDN	Electrobot USB-C 2S Charger	1	CC/CV charging and programming interface	Amazon

Layer	Component	Qty	Purpose	Supplier
PDN	5D-9 NTC Thermistor	1	Inrush current limiting	Robu.in
PDN	LM2596 5 V/3 A Buck Module	1	Brain rail voltage regulation	Amazon
PDN	4700 µF 16 V Electrolytic	1	Muscle rail bulk reservoir	Robu.in
PDN	1000 µF 16 V Electrolytic	1	Muscle rail reservoir	Robu.in
PDN	10 µF 16 V Tantalum	1	Brain rail output filter	Amazon
PDN	100 nF 50 V Ceramic	10+	MCU decoupling + rail filter	Amazon
PDN	18 AWG Silicone Wire	1 m	Muscle rail high-current paths	Local
PDN	28 AWG Signal Wire	1 m	Brain rail ground paths	Local

Appendix B — Complete Test Sketches

B.1 Test Sketch 1 — MOSFET Gate and Zone Verification

```
/*
 * P.H.A.N.T.O.M. System - Logic & MOSFET Verification
 * Testing Zones: D5, D6, D9, D10
 * Pattern: 3-Phase Attack (Burst -> Lull -> Spike)
 */

const int zones[] = {5, 6, 9, 10}; // The 4 MOSFET Gate Pins
const int numZones = 4;

void setup() {
  Serial.begin(115200);
  Serial.println("P.H.A.N.T.O.M. System Online.");
  Serial.println("Testing Zones D5, D6, D9, D10...");
  for (int i = 0; i < numZones; i++) {
    pinMode(zones[i], OUTPUT);
    digitalWrite(zones[i], LOW); // Ensure all are OFF initially
  }
}

void loop() {
  for (int z = 0; z < numZones; z++) {
    int currentPin = zones[z];
    Serial.print("Testing Zone on Pin: ");
    Serial.println(currentPin);

    // --- PHASE 1: THE BURST ---
    // Simulates filling the camera's temporal filter
    for (int i = 0; i < 4; i++) {
      digitalWrite(currentPin, HIGH);
      delay(20);
      digitalWrite(currentPin, LOW);
      delay(20);
    }

    // --- PHASE 2: THE LULL ---
    // Wait period to prevent full AGC recovery
    delay(75);

    // --- PHASE 3: THE SPIKE ---
    // The high-intensity disruption pulse
    digitalWrite(currentPin, HIGH);
    delay(15); // Visible 15ms for 5mm LED test
    digitalWrite(currentPin, LOW);

    // Short pause before moving to the next MOSFET
    delay(500);
  }
  Serial.println("All Zones Cycled. Restarting...");
  delay(1000);
}
```

B.2 Test Sketch 2 — Sensing Layer ADC Verification

```
/*
```

```
 * P.H.A.N.T.O.M. Initial Test: 5 Hunters + 1 Ambient
 * Used to verify the HPF/LPF stages across all sensors.
 */

// Define Pins for the 5 Hunter Channels
const int hunterPins[] = {A0, A1, A2, A3, A4};
const int ambientPin = A5; // The reference/ambient sensor

void setup() {
  Serial.begin(115200); // High speed for real-time plotting
  for (int i = 0; i < 5; i++) {
    pinMode(hunterPins[i], INPUT);
  }
  pinMode(ambientPin, INPUT);
  Serial.println("H1,H2,H3,H4,H5,Ambient"); // Headers for the Plotter
}

void loop() {
  // Read all 5 Hunter channels
  for (int i = 0; i < 5; i++) {
    int val = analogRead(hunterPins[i]);
    Serial.print(val);
    Serial.print(",");
  }
  // Read the Ambient channel
  int ambientVal = analogRead(ambientPin);
  Serial.println(ambientVal); // Final value with println to end the frame
  delay(10); // Small delay to stabilize readings
}
```